



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI MURUGAN AGENCIES UNIT II

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

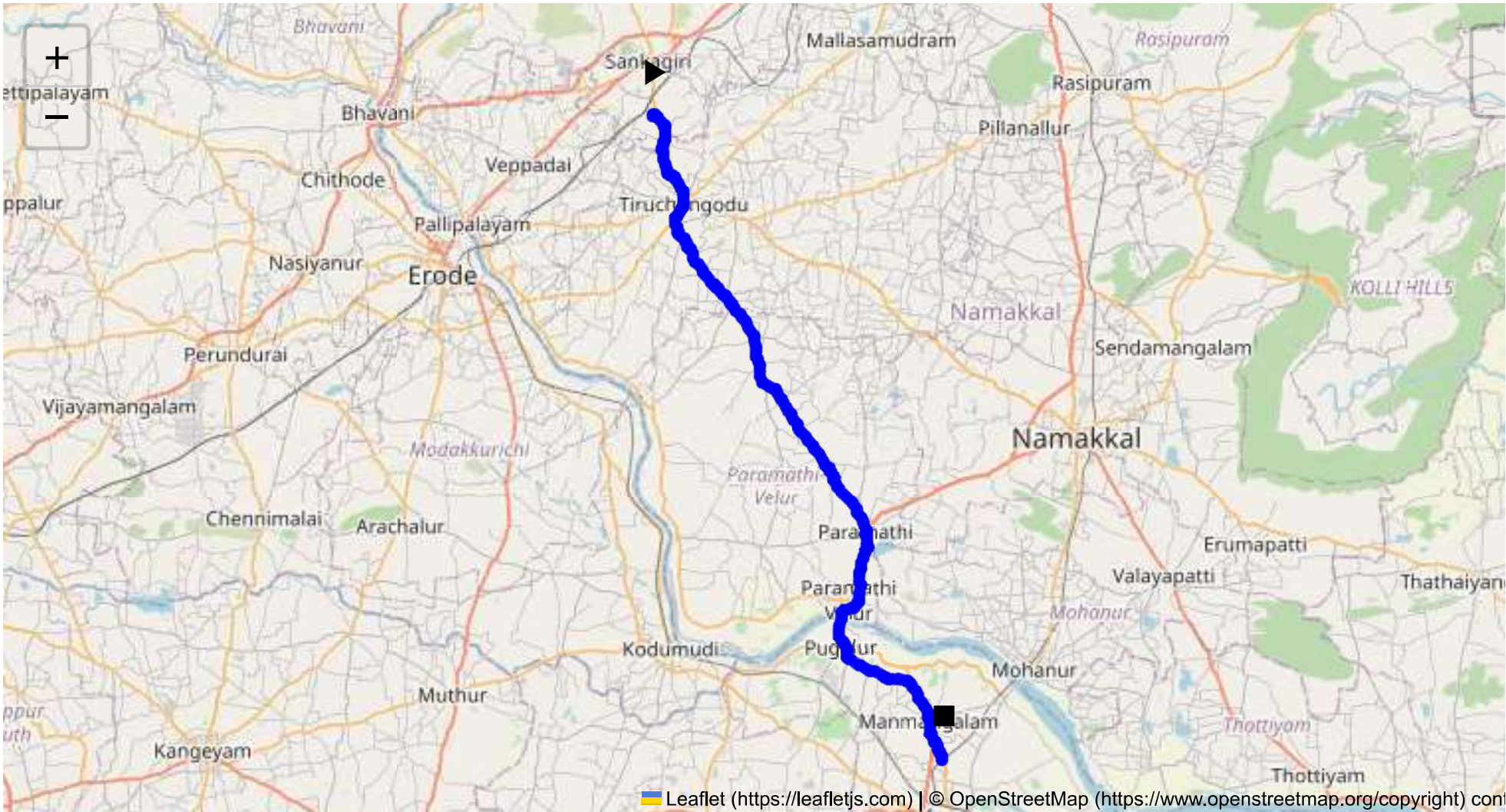
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 60.43 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.5 hours
Start: (11.4381, 77.8734)
End: (10.998904, 78.074713)



Welcome to the Journey Risk Management Study

**Route Safety Analysis from Sangagiri to Vennamalaipatty
Railway Bridge, Karur**

1. Overview of the Route Map

The route from Sangagiri to Vennamalaipatty Railway Bridge spans approximately 60.43 kilometers, primarily along major highways and internal roads. The journey mainly involves traversing the SH-86 and branching onto local roads closer to Karur.

2. Typical Weather Conditions and Potential Weather-related Hazards

This region of Tamil Nadu typically experiences a tropical climate. The primary weather concerns include:

- **Monsoon Season (June to September):** Significant rainfall can cause road flooding and reduced visibility.
- **Summer (March to May):** High temperatures can lead to heat-related vehicle issues such as tire blowouts.
- **Occasional Fog:** During winter months, primarily early morning fog can reduce visibility.

3. Analysis of Traffic Patterns

- **Peak Hours:** Congestion is common during morning (8:00-10:30 AM) and evening (5:00-7:30 PM) rush hours, particularly near urban centers and junctions.
- **Congestion-prone Areas:** Major intersections and market areas in villages like Sangagiri and the outskirts of Salem are notable for traffic buildup.

4. Assessment of Road Quality and Infrastructure

- **Highway Segments (SH-86):** Generally well-maintained with good signage but be cautious of occasional potholes.
- **Local Roads:** May have narrower lanes with variable conditions. Watch for livestock and unmarked speed bumps.

5. Suggestions for Alternative Routes for Emergencies

In emergencies, consider alternate routes via NH-544 or NH-81, although these may add distance and travel time. Always verify current road conditions through local traffic advisories.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Permits Required:** Ensure possession of requisite permits for transporting hazardous materials.
- **Restricted Hours:** Some regions may impose restrictions on heavy vehicle movement during peak urban hours.

7. Overview of Historical Incidents

There have been occurrences of rollovers and fires involving heavy vehicles near Salem due to high speeds and overtaking. Safety measures include reduced speed and heightened vigilance.

8. Environmental Considerations and Sensitive Areas

- **Protected Areas:** Avoid sensitive ecosystems near riverbeds, especially during monsoons when they are prone to flooding.
- **Agricultural Zones:** Awareness is needed to prevent contamination from hazardous spills.

9. Analysis of Communication Coverage

While urban and suburban areas generally have reliable communication networks, certain rural stretches may experience weak signals. Drivers should plan communication needs accordingly.

10. Estimated Emergency Response Times

- **Urban Segments:** Emergency services can arrive within 20-30 minutes.
- **Rural Areas:** Response times may extend to 45-60 minutes due to distance and road conditions.

11. Overall Summary of Risk Assessment

- **Risk Level:** Moderate. The route is generally safe for transportation with adherence to safety protocols and traveling during non-peak hours.
- **Recommendations:** Regular vehicle checks, adherence to speed limits, awareness of weather forecasts, and maintaining communication are crucial for safety.

By prioritizing these safety measures, drivers transporting hazardous materials can mitigate risks and ensure a smoother journey along this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Turn	High	11.37868, 77.89498	15 KM/Hr
5	Turn	High	11.37850, 77.89453	15 KM/Hr
6	Turn	Medium	11.16764, 78.01565	30 KM/Hr
7	Turn	High	11.15573, 78.02176	15 KM/Hr
8	Turn	High	11.14545, 78.02208	15 KM/Hr
9	Turn	High	11.14480, 78.02291	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Manmangalam	11.0275504, 78.0646437	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
19	school	Cheran Matriculation Higher Secondary School	10.9971306, 78.0743488	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16764, 78.01565



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.15573, 78.02176



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14480, 78.02291

